SQUAREFREE TRANSLATES AND ERDŐS PROBLEM #675

CLAUDE OPUS 5.5

Abstract. Erdős asked how large an integer $t \geq 1$ must be if, for every $1 \leq a \leq N$, a is squarefree if and only if $a+t$ is squarefree, and in particular whether necessarily $t > \exp(N^c)$ for some constant $c > 0$. We show that, for each fixed $0 < c < 25/72$ and all sufficiently large N, every such t is divisible by p^2 for every prime $p \leq N^c$; hence $t > \exp(N^c)$. The argument is local at each small prime: if $p^2 \nmid t$, a small squarefree number in a suitable residue class modulo p^2 or p yields a squarefree $a \leq N$ with $a+t$ not squarefree. The exponent $25/72$ comes from Nunes's bound for the least squarefree number in an arithmetic progression; a completely elementary bound already gives every $c < 1/4$. We also record the classical sieve argument showing that such t exist, in a form showing that the least one satisfies $\log t = O(N/\log N)$, and some numerical data.

About this version. This paper was written by Claude Opus 5.5 (Anthropic) in October 2026. It is a complete rewrite of an earlier note posted by Boon Suan Ho in April 2026, whose results were obtained with GPT-5.4 Pro. **The original version of this paper is available from `https://boonsuan.github.io/erdos675_squarefree_old.pdf`.** The main theorem and its proof are those of the original note. This version adds figures, numerical data, an account of the background and of later work, corrected attributions, and three short supplements: the elementary exponent $1/4$ (Theorem 1.2), an explicit upper bound for the least translate, and its refinement to $\log T(N) = O(N/\log N)$ (Proposition 1.3).

1. Introduction

1.1. **The problem.** In a 1979 collection of problems, Erdős [3, §5, p. 69] asked how long one must wait before the pattern of squarefree numbers on an initial interval recurs; this is part (c) of Erdős Problem #675 [1]. Write $\mathcal{S}$ for the set of squarefree positive integers and $\mathbf{1}_{\mathcal{S}}$ for its indicator function. For $N \geq 1$ call $t \geq 1$ *N-admissible* if

$$\mathbf{1}_{\mathcal{S}}(a+t) = \mathbf{1}_{\mathcal{S}}(a) \qquad (1 \leq a \leq N), \tag{1}$$

and let $T(N)$ be the least N-admissible integer. Figure 1 shows an example.

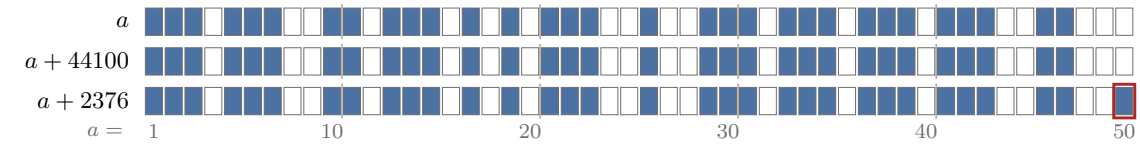


Figure 1. The squarefree indicator on $[1, 50]$ (top row; a filled cell means squarefree) and on the windows $[t+1, t+50]$ for $t = 44100 = 2^2 3^2 5^2 7^2$ and $t = 2376 = 2^3 3^3 11$; thin rules follow every tenth cell. The first window reproduces the pattern exactly; in fact $44100 = T(50)$. The second agrees for $a \leq 49$ (indeed $2376 = T(49)$) but fails at $a = 50$ (framed), where $50 = 2 \cdot 5^2$ is not squarefree while $2426 = 2 \cdot 1213$ is. The four small prime squares divide $T(50)$; Section 5.2 compares observed divisibility with the sufficient thresholds from Proposition 3.1.

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Erdős remarked that it is not hard to show that N-admissible integers exist for every N (Section 4 gives the classical argument), so $T(N)$ is well defined. He expected that $T(N) > \exp(N^c)$ for some constant $c > 0$, and was confident that $T(N)$ grows faster than any polynomial in N. The same paragraph of [3] asks two further questions of the same kind, about sums of two squares and about integers composed of primes from a prescribed set; we return to them, and to later work, in Section 5.3.

1.2. **The main result.** The main result of the original note [6] answers Erdős's question in the affirmative. It gives a divisibility condition on every admissible translate.

Theorem 1.1. *Let $0 < c < 25/72$. There is $N_0(c)$ such that for all $N \geq N_0(c)$, every N-admissible t satisfies*

$$\prod_{p \leq N^c} p^2 \,\Big|\, t.$$

Consequently $t \geq \exp(2\vartheta(N^c)) = \exp((2+o(1))N^c)$, where $\vartheta(x) = \sum_{p \leq x} \log p$; in particular $T(N) > \exp(N^c)$ for all large N.

Here and throughout, p and ℓ denote primes. The idea is as follows. Fix a prime p and suppose t is N-admissible but $p^2 \nmid t$. It suffices to find a *squarefree* $a \leq N$ with $p^2 \mid a + t$, for then $a + t$ is not squarefree, contradicting (1). There are two cases (Fig. 2).

- If $p \nmid t$, then $-t$ is a reduced residue class modulo p^2, and we take a to be the least squarefree number in that class.
- If $p \parallel t$, write $t = pu$ with $p \nmid u$. Now every $a \equiv -t \pmod{p^2}$ is divisible by p, so we take $a = pb$ with b the least squarefree number $\equiv -u \pmod{p}$. Then a is squarefree (as $p \nmid b$) and $a + t = p(b + u)$ is divisible by p^2.

Everything therefore reduces to how large the least squarefree number in a reduced class modulo q can be, for $q = p^2$ and $q = p$. If it is at most Cq^κ with $\kappa \geq 1$, both witnesses are at most $Cp^{2\kappa}$, so $p^2 \mid t$ for every $p \leq (N/C)^{1/(2\kappa)}$; multiplying over these primes gives $\log t \geq 2\vartheta\big((N/C)^{1/(2\kappa)}\big)$. The input is a bound of Nunes [11], refining Heath-Brown [5], with $\kappa = 36/25 + \varepsilon$ (Section 2); this gives every $c < 25/72$. The proof never uses that admissible t exist.

1.3. **Further results.** An elementary bound for the least squarefree number in a progression, with $\kappa = 2$, already gives a positive exponent:

Theorem 1.2. *For every $N \geq 1$, every N-admissible t is divisible by p^2 for every prime $p \leq (N/9)^{1/4}$. Consequently, for every $c < 1/4$, $T(N) > \exp(N^c)$ for all large N.*

So the qualitative answer to Erdős's question needs nothing beyond the Chinese remainder theorem and Chebyshev's estimate; the deeper input in Theorem 1.1 improves the exponent from $1/4$ to $25/72 \approx 0.347$. In the other direction, the sieve argument that produces admissible translates gives an exponential upper bound, and a more careful choice of the sieving modulus saves a logarithm.

Proposition 1.3.

(i) *For every $N \geq 4$, $T(N) \leq 3N^4 \prod_{p \leq N} p^4 = \exp((4+o(1))N)$.*
(ii) *For $N \geq 2$, $\log T(N) \ll N/\log N$, with an absolute implied constant.*

Thus $N^{25/72-o(1)} \leq \log T(N) \ll N/\log N$. We do not know the true order of $\log T(N)$; Section 5 discusses the limits of the method and some numerical data.

Theorem 1.1 and its proof are from Ho's note [6]. Theorem 1.2 and Proposition 1.3 are supplements added in this version: part (i) of the proposition is the classical sieve argument made explicit, and part (ii) refines it.

2. The least squarefree number in a residue class

For $q \geq 1$ and an integer r with $(r, q) = 1$, let $L(q, r)$ denote the least positive squarefree integer congruent to r modulo q. Since the values $L(q, r)$ for the $\varphi(q)$ reduced classes are distinct, $\max_r L(q, r) \geq \varphi(q)$. Thus a bound $L(q, r) \ll q^{\kappa}$, uniform over all moduli and reduced classes, cannot hold with a fixed $\kappa < 1$.

An elementary counting argument gives the bound $L(q, r) < 9q^2$; Theorem 1.2 rests on it.

Lemma 2.1. *For all $q \geq 1$ and $(r, q) = 1$ we have $L(q, r) < 9q^2$.*

Proof. Let $X = 9q^2$. The class r mod q contains exactly $X/q = 9q$ integers in $[1, X]$, all coprime to q. If such an m is not squarefree, then $\ell^2 \mid m$ for some prime $\ell \leq \sqrt{X} = 3q$ with $\ell \nmid q$. For a fixed such ℓ, the conditions $m \equiv r \pmod q$ and $\ell^2 \mid m$ define one class modulo $q\ell^2$, which has at most $X/(q\ell^2) + 1$ elements in $[1, X]$. Since $(2j + 1)^2 > 4j(j + 1)$,

$$\sum_{\ell} \frac{1}{\ell^2} \;<\; \frac{1}{4} + \sum_{j \geq 1} \frac{1}{(2j+1)^2} \;<\; \frac{1}{4} + \frac{1}{4} \sum_{j \geq 1} \frac{1}{j(j+1)} \;=\; \frac{1}{2}.$$

Writing $\pi(x)$ for the number of primes at most x, we also have $\pi(3q) \leq (3q + 1)/2$, since every prime other than 2 is odd and at least 3. Hence the number of non-squarefree elements is less than $\frac{1}{2} \cdot 9q + \frac{3q+1}{2} = 6q + \frac{1}{2} < 9q$. So the class contains a squarefree $m \leq X$, and $m \neq X$ because $X = 9q^2$ is not squarefree. □

Improving the exponent 2 is a classical problem. Counting squarefree numbers in progressions with an error term, Prachar [12] obtained the exponent $3/2 + \varepsilon$; Heath-Brown [5] reached $13/9 + \varepsilon$. For a bound uniform over all moduli, we use the following theorem of Nunes.

Theorem 2.2 (Nunes [11, Cor. 1.2])**.** *For every $\varepsilon > 0$ there is C_ε such that $L(q, r) \leq C_\varepsilon\, q^{36/25+\varepsilon}$ for all $q \geq 1$ and all r with $(r, q) = 1$.*

Nunes deduces this from an asymptotic formula [11, Thm. 1.1]. Write μ for the Möbius function, so that $\mu^2 = \mathbf{1}_{\mathcal{S}}$. For each $\eta > 0$ there is $\delta = \delta(\eta) > 0$ such that

$$\sum_{\substack{n \leq X \\ n \equiv r \pmod q}} \mu^2(n) = \frac{1}{\varphi(q)} \sum_{\substack{n \leq X \\ (n,q)=1}} \mu^2(n) + O_\eta\Big(\frac{X^{1-\delta}}{q}\Big),$$

uniformly for $q \leq X^{25/36-\eta}$ and $(r, q) = 1$; that is, $25/36$ is an exponent of distribution for the squarefree numbers. The main term is $\gg X/q$. Given $\varepsilon > 0$, choose $\eta > 0$ so that $(36/25 + \varepsilon)(25/36 - \eta) > 1$, and take $X = q^{36/25+\varepsilon}$. Then $q \leq X^{25/36-\eta}$, and the error term is smaller than the main term once q is large, so the class r mod q contains a squarefree number at most X. This proves Theorem 2.2 for large q; the remaining moduli are absorbed into C_ε. In the same way an exponent of distribution β for a family of moduli gives $\kappa = 1/\beta + \varepsilon$ for that family.

We use Theorem 2.2 only for $q = p$ and $q = p^2$. The arXiv preprint of Nunes's paper [10] states its Theorem 1.1 and Corollary 1.2 for squarefree moduli only, which covers $q = p$ but not $q = p^2$. The published version [11] states them for all moduli; Mangerel [8, §1], citing the published version, likewise describes $25/36$ as the best exponent of distribution available for all moduli. Even without the case $q = p^2$ of Theorem 2.2, Heath-Brown's bound for all q together with the argument below would give Theorem 1.1 for $c < 9/26 \approx 0.346$ (see Table 1).

3. The local argument

The heart of the matter is the following criterion. For a prime p put

$$N_p := \max\Big\{ \max_{r \bmod p^2,\ p\nmid r} L(p^2, r),\quad p \cdot \max_{r \bmod p,\ p\nmid r} L(p, r) \Big\}.$$

This is a sufficient threshold, obtained by taking the larger of the two witness bounds in the proof below; it need not be the least N from which on $p^2 \mid t$ is forced (see Section 5.2).

Proposition 3.1. *Let p be prime and $N \ge N_p$. Then $p^2 \mid t$ for every N-admissible t.*

Proof. Let t be N-admissible and suppose $p^2 \nmid t$. We produce a squarefree $a \in [1, N]$ with $p^2 \mid a + t$; then $a + t$ is not squarefree, contradicting (1).

Case 1: $p \nmid t$. The class $-t \bmod p^2$ is reduced. Let $a = L(p^2, -t)$. Then a is squarefree, $a \le N_p \le N$, and $a + t \equiv 0 \pmod{p^2}$.

Case 2: $p \mid t$ but $p^2 \nmid t$. Write $t = pu$ with $p \nmid u$, let $b = L(p, -u)$, and put $a = pb$. Since $b \equiv -u \not\equiv 0 \pmod p$, the prime p does not divide the squarefree number b, so a is squarefree. Moreover $a \le N_p \le N$, and $a + t = p(b + u)$ with $p \mid b + u$, so $p^2 \mid a + t$. □

Figure 2 shows the two cases for $p = 5$.

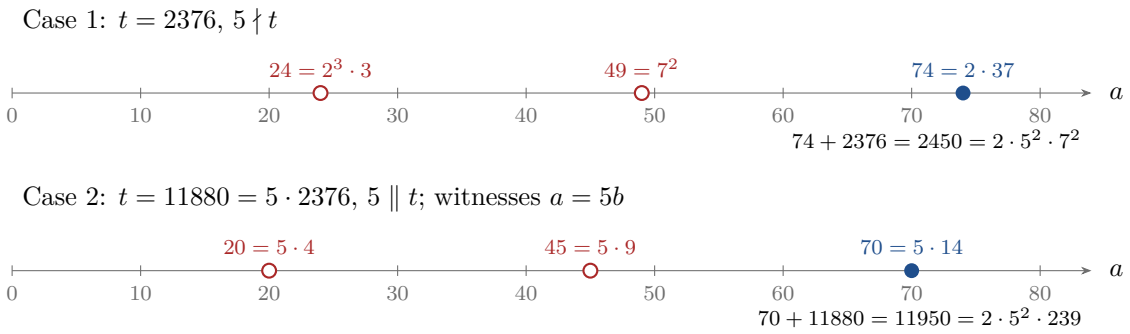


Figure 2. The two cases of Proposition 3.1 for $p = 5$. Top: $5 \nmid t$, so we look along the class $a \equiv -t \equiv 24 \pmod{25}$; its first two members 24 and 49 are not squarefree (open circles), and the witness is $a = L(25, 24) = 74$. Bottom: $5 \parallel t$, $t = 5u$ with $u = 2376 \equiv 1 \pmod 5$; we look at $a = 5b$ with $b \equiv -u \equiv 4 \pmod 5$, i.e. $a \equiv 20 \pmod{25}$; $b = 4 = 2^2$ and $b = 9 = 3^2$ are not squarefree, and the witness is $b = 14$, $a = 70$. In both cases a is squarefree and $25 \mid a + t$, so neither t is N-admissible once $N \ge 74$, resp. $N \ge 70$. These witnesses need not be the first mismatches: Fig. 1 shows that $t = 2376$ already fails at $a = 50$. Checking all reduced classes gives $\max_r L(25, r) = 74$ and $\max_r L(5, r) = 14$, hence $N_5 = \max\{74, 5 \cdot 14\} = 74$.

To compute N_p, enumerate the squarefree integers in increasing order until every reduced class modulo p and modulo p^2 has appeared, then take the two maxima in the definition. For instance $N_2 = 3$, $N_3 = 17$, $N_5 = 74$, $N_7 = 174$, $N_{11} = 417$, $N_{13} = 583$. Any upper bound for $L(q, r)$ at $q = p$ and $q = p^2$ bounds N_p:

Corollary 3.2. *Suppose that $C \ge 1$ and $\kappa \ge 1$ are such that $L(q, r) \le C q^\kappa$ whenever $q \in \{p, p^2\}$ for a prime p and $(r, q) = 1$. Then $N_p \le C p^{2\kappa}$ for every prime p. Hence for every $N \ge 1$, every N-admissible t is divisible by p^2 for all primes $p \le (N/C)^{1/(2\kappa)}$, and so $\log t \ge 2\vartheta\big((N/C)^{1/(2\kappa)}\big)$.*

Proof. The first maximum in the definition of N_p is at most $C(p^2)^\kappa = C p^{2\kappa}$, and the second term is at most $p \cdot C p^\kappa \le C p^{2\kappa}$ because $\kappa \ge 1$. The rest follows from Proposition 3.1, since the p^2 are pairwise coprime. □

3.1. **Proofs of the lower bounds.** We use Chebyshev's estimate $\vartheta(x) \gg x$ for $x \geq 2$ [4, Ch. XXII]; the prime number theorem $\vartheta(x) \sim x$ is needed only for the constant $2 + o(1)$ in Theorem 1.1.

Proof of Theorem 1.2. By Lemma 2.1 the hypothesis of Corollary 3.2 holds with $C = 9$ and $\kappa = 2$, which gives the divisibility statement. For the bound, given $c < 1/4$ we have $\log T(N) \geq 2\vartheta((N/9)^{1/4}) \gg N^{1/4}$, which exceeds N^c for all large N. □

Proof of Theorem 1.1. Fix $0 < c < 25/72$ and choose $\varepsilon > 0$ so small that $c(36/25 + \varepsilon) < 1/2$. By Theorem 2.2 the hypothesis of Corollary 3.2 holds with $\kappa = 36/25 + \varepsilon$ and $C = C_\varepsilon$. For prime $p \leq N^c$ we then have $Cp^{2\kappa} \leq CN^{2c\kappa} \leq N$ as soon as $N^{1-2c\kappa} \geq C$, which holds for $N \geq N_0(c)$. For such N, Corollary 3.2 gives $p^2 \mid t$ for all $p \leq N^c$, hence $\prod_{p \leq N^c} p^2 \mid t$ and $\log t \geq 2\vartheta(N^c)$. By the prime number theorem $2\vartheta(N^c) = (2 + o(1))N^c$, which exceeds N^c for large N. (Without the prime number theorem: apply the divisibility with some $c' \in (c, 25/72)$ in place of c and use $\vartheta(N^{c'}) \gg N^{c'}$.) □

3.2. **Other exponents.** If the bounds for $q = p^2$ and $q = p$ have different exponents κ_2 and κ_1, the same argument gives the range $p \leq N^c$ for every $c < \min\{1/(2\kappa_2),\, 1/(1 + \kappa_1)\}$. With $\kappa_1 = \kappa_2 = \kappa \geq 1$ the first term is no larger than the second, since $2\kappa \geq 1 + \kappa$, and the two coincide only at $\kappa = 1$; so the modulus p^2 (Case 1) is the bottleneck. Table 1 lists the resulting ranges for the known values of κ.

input	κ	$1/(2\kappa)$ (modulus p^2)	$1/(1+\kappa)$ (modulus p)
Lemma 2.1	2	$1/4 = 0.250$	$1/3 \approx 0.333$
Prachar [12]	3/2	$1/3 \approx 0.333$	$2/5 = 0.400$
Heath-Brown [5]	13/9	$9/26 \approx 0.346$	$9/22 \approx 0.409$
Nunes [11]	36/25	$25/72 \approx 0.347$	$25/61 \approx 0.410$
conjectural	1	1/2	1/2

Table 1. Ranges obtained from Corollary 3.2. If $L(q, r) \ll_\varepsilon q^{\kappa+\varepsilon}$ for $q \in \{p, p^2\}$, then for each fixed $c < \min\{1/(2\kappa), 1/(1 + \kappa)\} = 1/(2\kappa)$ and all large N, every N-admissible t is divisible by p^2 for all $p \leq N^c$. The last column never gives the binding restriction.

4. Upper bounds: the translation property

A set $A \subseteq \mathbb{N}$ has the *translation property* if for every n there is an integer $t \geq 1$ such that

$$a \in A \iff a + t \in A \qquad (1 \leq a \leq n). \tag{2}$$

For $A = \mathcal{S}$ this says that N-admissible integers exist for every N. That they do is the classical part of the problem. Erdős [3, §5] notes that this is not hard to show, and more generally that the sieve of Eratosthenes gives the translation property for the integers divisible by none of $b_1 < b_2 < \cdots$, whenever the b_i are pairwise coprime and $\sum_i 1/b_i < \infty$; the squarefree numbers are the case $\{b_i\} = \{p^2\}$. The argument is the standard one for showing that several linear forms take squarefree values simultaneously, a question whose quantitative study goes back to Mirsky [9], who counted $n \leq x$ with $n + h_1, \ldots, n + h_s$ all r-free. We give it in a form that produces the explicit bound of Proposition 1.3 (i), and then refine it.

Proof of Proposition 1.3 (i). Let $N \geq 4$, let $P = \prod_{p \leq N} p^2$, and let B be the set of squarefree $a \in [1, N]$, with $k = |B|$. Since $4 \notin B$ we have $k \leq N - 1$.

We look for t of the form $t = Pm$ with $m \geq 1$. Every such t handles the non-squarefree $a \leq N$ automatically: if $p^2 \mid a$ then $p \leq N$, so $p^2 \mid P \mid t$ and $p^2 \mid a + t$. It therefore suffices to find m with $Pm + a$ squarefree for all $a \in B$. For $a \in B$ and $p \leq N$ we have $Pm + a \equiv a \not\equiv 0 \pmod{p^2}$, so only primes $\ell > N$ can obstruct.

Let X be a positive integer and call $m \in [1, X]$ *bad* if $\ell^2 \mid Pm + a$ for some $a \in B$ and some prime $\ell > N$. Such ℓ satisfies $\ell^2 \leq PX + N \leq 2PX$. For fixed a and ℓ, since $\ell \nmid P$, the condition $\ell^2 \mid Pm + a$ fixes m modulo ℓ^2, so it holds for at most $X/\ell^2 + 1$ values of $m \leq X$. As $\sum_{\ell > N} \ell^{-2} < \sum_{n > N} n^{-2} < 1/N$, the number of bad m is less than

$$k\Big(\frac{X}{N} + \sqrt{2PX}\Big) \;\leq\; X - \frac{X}{N} + N\sqrt{2PX}.$$

This is less than X once $X > 2N^4P$. Taking $X = 2N^4P + 1$ we find a good $m \leq X$, and then $t = Pm \leq P(2N^4P + 1) \leq 3N^4P^2$ is N-admissible. Finally $P^2 = \exp(4\vartheta(N)) = \exp((4 + o(1))N)$. □

The same count shows that, for fixed N, at least a proportion $1 - k/N - o(1) \geq 1/N - o(1)$ of the $m \leq X$ give admissible translates Pm, as $X \to \infty$.

The proof of (i) bounds $\sum_{\ell > N} \ell^{-2}$ by $1/N$, whereas, since only primes occur, its true order is $1/(N \log N)$. Using this, it suffices to put into P the squares of the primes up to a constant multiple of $N/\log N$ (the non-squarefree $a \leq N$ only need the primes up to $\sqrt{N}$), which gives (ii).

Proof of Proposition 1.3 (ii). By Chebyshev's estimate [4, Thm. 7] there is an absolute constant $C_0 \geq 1$ with $\pi(x) \leq C_0 x/\log x$ for $x \geq 2$, where $\pi(x)$ is the number of primes up to x. Splitting into dyadic ranges, for $y \geq 2$,

$$\sum_{\ell > y} \frac{1}{\ell^2} \;\leq\; \sum_{j \geq 0} \frac{\pi(2^{j+1}y)}{(2^j y)^2} \;\leq\; \sum_{j \geq 0} \frac{C_0\, 2^{j+1} y}{4^j y^2 \log y} \;=\; \frac{4C_0}{y \log y}.$$

Let $N \geq 2$, put $y = 16C_0 N/\log N$, and let $P = \prod_{p \leq y} p^2$. Since $\log N < \sqrt{N}$, we have $y \geq 16\sqrt{N}$, so $y \geq 2$, $y \geq \sqrt{N}$ and $\log y \geq \frac{1}{2}\log N$, hence $y \log y \geq 8C_0 N$ and

$$N \sum_{\ell > y} \frac{1}{\ell^2} \;\leq\; \frac{4C_0 N}{8C_0 N} \;=\; \frac{1}{2}.$$

Now repeat the proof of (i) with this P. If $a \leq N$ is not squarefree then $p^2 \mid a$ for some $p \leq \sqrt{N} \leq y$, so $p^2 \mid P$ and $p^2 \mid a + Pm$. For $a \in B$ and $p \leq y$ we have $Pm + a \equiv a \not\equiv 0 \pmod{p^2}$, so $m \in [1, X]$ is bad only if $\ell^2 \mid Pm + a$ for some $a \in B$ and some prime $\ell > y$. Such ℓ satisfy $\ell^2 \leq PX + N \leq 2PX$, and for fixed a and ℓ (note $\ell \nmid P$) at most $X/\ell^2 + 1$ values of $m \leq X$ qualify. Since $k = |B| \leq N$, the number of bad m is at most

$$NX \sum_{\ell > y} \frac{1}{\ell^2} + N\sqrt{2PX} \;\leq\; \frac{X}{2} + N\sqrt{2PX}.$$

Take $X = 8N^2P + 1$. Then $X > 8N^2P$ gives $N\sqrt{2PX} < X/2$, so fewer than X values of $m \leq X$ are bad, and a good $m \leq X$ yields the N-admissible translate $t = Pm \leq P(8N^2P + 1) \leq 9N^2P^2$. Therefore

$$\log T(N) \;\leq\; 4\vartheta(y) + \log(9N^2) \;\ll\; y \;\ll\; \frac{N}{\log N},$$

using $\vartheta(y) \ll y$. □

5. Remarks

5.1. **Limits of the method.** Put $D(N) = 2\sum_{p:\,N_p \le N} \log p$. Proposition 3.1 gives $\log T(N) \ge D(N)$. Since $\max_r L(p^2, r) \ge \varphi(p^2) = p^2 - p$, we always have $N_p \ge p^2 - p$, so $N_p \le N$ forces $p \le \sqrt{N} + 1$ and

$$D(N) \;\le\; 2\vartheta(\sqrt{N} + 1) \;=\; (2 + o(1))\sqrt{N}.$$

Thus the criterion $N \ge N_p$, applied independently at each prime, cannot yield a lower-bound exponent greater than $1/2$. This is a coarse ceiling for this uniform criterion, not a claim that a lower bound of order $\sqrt{N}$ is attained, nor a limit on every possible use of local congruences. The conjectural bound $L(q, r) \ll q^{1+\varepsilon}$ (which would follow from the expected exponent of distribution 1 for squarefree numbers; see [10, §1]) would reach this ceiling, giving every $c < 1/2$ in Theorem 1.1.

Numerically N_p is close to p^2. Figure 3 shows N_p/p^2 for all primes $p < 400$; the ratio stays below 9 (its largest value, about 8.83, occurs at $p = 353$), and for every one of these primes the first maximum in the definition of N_p (Case 1) is the larger.

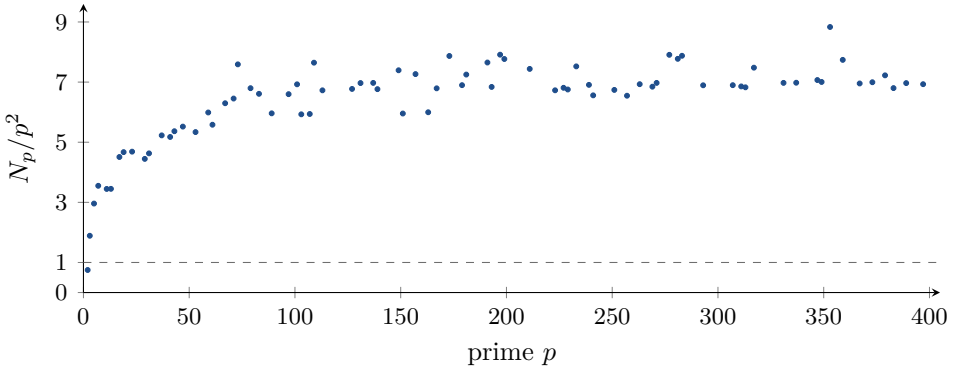


Figure 3. The sufficient threshold N_p of Proposition 3.1, normalised by p^2, for all primes $p < 400$ (computed exactly by this enumeration); the dashed line marks $N_p = p^2$. For $N \ge N_p$ every N-admissible t is divisible by p^2. The proven bounds give only $N_p \le 9p^4$ (Lemma 2.1) and $N_p \ll p^{72/25+\varepsilon}$ (Theorem 2.2); these finite data are compatible with $N_p = p^{2+o(1)}$, the size predicted by the conjectured exponent 1 for the least squarefree number in a progression.

5.2. **Numerical data.** Table 2 lists $T(N)$ for all $N \le 288$. It was found by scanning $t = 1, 2, \ldots, 10^{11}$ in increasing order by computer and recording, for each length n, the first t whose window $[t+1, t+n]$ agrees with $[1, n]$ in the sense of (1); because the scan is in increasing order, these are the least translates. No $t \le 10^{11}$ agrees on $[1, 289]$, so $T(289) > 10^{11}$. The divisibility forced by Proposition 3.1 is visible: $4 \mid T(N)$ for $N \ge N_2 = 3$ and $9 \mid T(N)$ for $N \ge N_3 = 17$. In the data, $p^2 \mid T(N)$ for $p \in \{5, 7\}$ already once $N \ge 50$, well before the sufficient thresholds $N_5 = 74$ and $N_7 = 174$; on the other hand 11^2 divides $T(N)$ for $161 \le N \le 168$ but not for $169 \le N \le 241$, so below N_p the divisibility need not persist. The range is far too small to say anything about the asymptotic growth of $\log T(N)$.

N	$T(N)$	factorisation	N	$T(N)$	factorisation
1–2	1	1	65–104	88 200	$2^3 \cdot 3^2 \cdot 5^2 \cdot 7^2$
3–4	4	2^2	105–109	485 100	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 11$
5	12	$2^2 \cdot 3$	110–118	573 300	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 13$
6–8	28	$2^2 \cdot 7$	119–120	1 455 300	$2^2 \cdot 3^3 \cdot 5^2 \cdot 7^2 \cdot 11$
9–12	36	$2^2 \cdot 3^2$	121–160	25 886 700	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 587$
13–24	180	$2^2 \cdot 3^2 \cdot 5$	161–168	37 352 700	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^3 \cdot 11^2$
25–30	900	$2^2 \cdot 3^2 \cdot 5^2$	169–185	450 481 500	$2^2 \cdot 3^4 \cdot 5^3 \cdot 7^2 \cdot 227$
31–49	2 376	$2^3 \cdot 3^3 \cdot 11$	186–241	3 609 452 700	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 81847$
50–64	44 100	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2$	242–288	4 108 797 000	$2^3 \cdot 3^2 \cdot 5^3 \cdot 7^3 \cdot 11^3$

Table 2. The least admissible translates $T(N)$ for $N \le 288$, with their factorisations; read down the left block, then down the right. Each row covers a range of N on which $T(N)$ is constant. Powers 11^e with $e \ge 2$ are shown in red. The search also gives $T(289) > 10^{11}$.

5.3. **Later work and open questions.** Besides question (c) on the squarefree numbers, Erdős [3, §5] asked two further questions about the translation property (2) (labels as in [1]):

(a) Does the set of sums of two squares have the translation property?
(b) Partition the primes as $P \sqcup Q$ with each part containing $\gg x/\log x$ primes up to x for all large x. Can the set of integers composed only of primes from P have the translation property?

He also noted that Brun's method gives the translation property for the integers divisible by none of a pairwise coprime sequence $b_1 < b_2 < \cdots$ under the weaker condition $\sum_{b_i<x} 1/b_i = o(\log\log x)$, and that he had no non-trivial result if this condition is dropped as well.

Yu Leon Liu (`leon2k2k2k`) adapted the local argument to question (a) [7]. Let E be the set of sums of two squares; a positive integer lies in E if and only if every prime $p \equiv 3 \pmod 4$ divides it to an even power. Suppose t satisfies (2) with $A = E$, and let $p \equiv 3 \pmod 4$ be a prime with $p \nmid t$. Choose two distinct lifts $r_0 \not\equiv r_1 \pmod{p^2}$ of the class $-t \bmod p$, combine each with 1 mod 4 by the Chinese remainder theorem, and apply Linnik's theorem to the resulting reduced classes modulo $4p^2$: this gives primes $\ell_i \equiv 1 \pmod 4$ with $\ell_i \equiv r_i \pmod{p^2}$ and $\ell_i \le C(4p^2)^L$, where C and L are absolute constants. Suppose $\ell_0, \ell_1 \le n$; this holds for all $p \le n^c$ provided $2Lc < 1$ and n is sufficiently large. Each ℓ_i lies in E, so the translation property gives $\ell_i + t \in E$. Since $\ell_i \equiv -t \pmod p$, we have $p \mid \ell_i + t$; as p occurs to an even exponent in every element of E, we get $p^2 \mid \ell_i + t$, so $\ell_0 \equiv -t \equiv \ell_1 \pmod{p^2}$, a contradiction. Hence every prime $p \equiv 3 \pmod 4$ with $p \le n^c$ divides t, and the prime number theorem for the progression 3 mod 4 gives $\log t \ge \sum_{p \le n^c,\, p \equiv 3\,(4)} \log p = (\tfrac{1}{2} + o(1))n^c$. Given $c < 1/(2L)$, applying this with some $c' \in (c, 1/(2L))$ in place of c gives $t > \exp(n^c)$ for all large n. With Xylouris's value $L \le 5$ [13] this holds for every $c < 1/10$. This bounds the size of such t but does not decide whether they exist.

A forum user (`aditya`) suggested improving 25/72 to 3/8 using a result of Zhong and Zhang [14] on squarefree numbers in progressions to prime-power moduli $q = p^n$, on the grounds that it goes beyond the exponent of distribution 3/4; in the notation of Table 1, an exponent of distribution β for the moduli p^2 would give $c < \min\{\beta/2,\, 25/61\}$. Liu pointed out on the forum [2] that Zhong and Zhang's theorem requires $n \ge 4$ and their corollary $n \ge 5$, whereas the argument needs $n = 2$. We were unable to consult the full text and know of no result covering $q = p^2$, so we regard the suggested improvement as unestablished.

Questions (a) and (b) remain open, as does Erdős's question about pairwise coprime b_i when the condition $\sum_{b_i<x} 1/b_i = o(\log\log x)$ is dropped. For (c), the order of growth of $\log T(N)$

remains open: it lies between $N^{25/72-o(1)}$ and $O(N/\log N)$, and the criterion of Proposition 3.1, applied prime by prime, cannot give more than $(2+o(1))\sqrt{N}$.

Acknowledgements and provenance

The local divisibility argument and Theorem 1.1 were found by GPT-5.4 Pro, prompted by Boon Suan Ho, and appeared in Ho's April 2026 note [6]. On the problem's forum thread [2], Nat Sothanaphan reported that a standard check of that note found no issues and Terence Tao endorsed it; the later comments by Yu Leon Liu (`leon2k2k2k`) and `aditya` described in Section 5.3 are on the same thread. This text, including Theorem 1.2, both parts of Proposition 1.3, and the computations, was written by Claude Opus 5.5.

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